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Chapter 1

Introduction

This chapter provides an overview of the research on Alzheimer's disease detection using brain MRI images and machine learning and deep learning techniques. It presents the introduction, motivation, objectives, methodology, research outcomes, and overall organization of the study.

1.1 Introduction

The primary cause of dementia, Alzheimer's disease (AD) is a progressive neurological illness marked by a slow deterioration in memory, cognitive function, and day to day functioning. It mainly impacts the elderly and is linked to the buildup of tau protein tangles and amyloid beta plaques in the brain, which causes neural degeneration [1]. Since the early signs of Alzheimer's disease are frequently weak and difficult to differentiate from regular aging, early detection of the illness continues to be a major issue in clinical practice [2]. Using the general population data, clinical documents, and neuroimaging datasets, recent developments in artificial intelligence, especially machine learning (ML) and deep learning (DL), have shown impressive promise in enhancing the early detection and forecasting of Alzheimer's disease [3]. Research from 2023 to 2025 shows that while deep learning predicts are better at recognizing intricate patterns and attaining greater accuracy, traditional machine learning models like Support Vector Machines, Random Forest, and Logistic Regression offer dependable classification performance [4]. Additionally, the creation of scalable and affordable diagnostic platforms which are particularly helpful in poor-resource healthcare settings has been made possible by the integration of public health information and electronic health records [5]. Thus, a possible method for early the condition known as Alzheimer's detection is to combine machine learning with deep learning techniques, which will ultimately improve patient care and lessen the burden of healthcare worldwide.

1.2 Motivation

According to the most prevalent neurological conditions in the world, Alzheimer's disease affects millions of individuals and results in gradual loss of memory, cognitive decline, and a worse quality of life. For prompt healthcare and to halt

the course of the disease, early diagnosis is essential. But conventional diagnostic methods are frequently expensive, laborious, and rely on professional interpretation. New possibilities for automated and precise illness diagnosis have been made possible by recent developments in artificial intelligence, notably machine learning and deep. The desire to create an effective MRI based Alzheimer's disease detection technique that facilitates quicker diagnosis, better healthcare assistance, and improved medical decisions is the driving force behind this research.

1.3 Objectives

The central purpose of the research is to design a model that will be efficient in classifying MRI images for early detection of Alzheimer's disease through machine learning and deep learning. The study aims to use public health data in designing an effective detection system and employ machine learning methods through two binary classification systems. In addition, the study seeks to design and employ deep learning and customized models to increase prediction accuracy. Moreover, the study seeks to evaluate and compare the efficiency of the models designed through the performance metrics of accuracy, precision, recall, and F1 score.

1.4 Methodology

The suggested technology uses MRI scans to diagnose Alzheimer's disease at many stages. First, 12,800 MRI pictures from four classifications Mild Demented, Moderate Demented, Non-Demented, and Very Mild Demented were gathered from Kaggle. To enhance data quality and model generalization, the photos were preprocessed using resizing, normalization, scaling, and data augmentation. DenseNet201 and a modified Inception Resnetv2 combined with customize layers were used and assessed for deep learning categorization. Four different classes were then divided into two groups for machine learning tests: Alzheimer's disease (AD) and non-AD. Lastly, conventional evaluation criteria were used to evaluate the performance of Support Vector Machine (SVM)

and Quantum Support Vector Machine (QSVM) models after they had been trained and tested.

1.5 Project Outcome

- Created a systemic method for detecting Alzheimer's disease utilizing MRI brain pictures combining both machine learning and deep learning techniques.
- Densenet201 and a modified Inception Resnetv2 combined with customize layers for multi-class Alzheimer's disorder classification have been successfully utilized.
- For four-class classifying, use Densenet201 and the Inception Resnetv2 models.
- SVM and QSVM algorithms were used after the dataset was divided into binary groups (AD and Non-AD).
- Shown how early signs of Alzheimer's identification and healthcare may be aided by artificial intelligence-supported diagnostic technologies.

1.6 Organization of the Report

Chapter 1: The first introduction to the brain MRI Disease Methodologies part of this chapter included a summary of the study's objectives, purpose, goal, and expected outcomes. The whole scope of the study is explained here.

Chapter 2: The present section addresses the special accomplishments in the domain of brain MRI illness thus far. Assessments of the existing literature, scholarly comparisons, and associated publications considered to be pertinent to this subject are all included in the present chapter.

Chapter 3: The overall ultimate brain MRI illness approach employed in this study is explained in this chapter. This section covers a number of subjects, including the research specifics and methodology, suitable models, data

collection, data analysis, and both functional and non-functional needs requirements that need to be addressed while attempting to develop this project.

Chapter 4: The simulation findings, the evaluation of performance, and results analysis are presented in this chapter. The achievement of each paradigm is discussed in this chapter. This section assesses our methodology for various deep learning & machine learning methods, including custom models, utilizing assessment metrics like as precision, recall, accuracy, a roc curve, etc.

Chapter 5: This chapter offered an overview of the study, information about forthcoming events, and all future research on the results of brain MRI scans images. This chapter discusses the effects on the environment and other aspects of life.

Chapter 6: The last section of the chapter addresses the limitations of this job that might impact future employees with comparable goals.

Chapter 2

Background

This chapter presents the background and previous research related to Alzheimer's disease detection using brain MRI images and machine learning and deep learning techniques. It discusses the relevant literature, comparative analysis, research gaps, and a summary of existing studies that support the proposed system.

2.1 Introduction

This section discusses the particular accomplishments made in this field thus far. This chapter includes evaluations of relevant literature, comparative analysis, and linked publications. The primary subjects of this part are the research findings, methodology, review of the literature, and studies summary. After providing an overview of the appropriate literature in the section focusing on characteristics, we will address the articles, their methodology, and the validity of other scholarly sources that are relevant to our research in the following section on comparison research methodologies.

2.2 Literature Review

It has been a difficult issue to classify MRI imaging utilizing many influenced conditions. An overview of different methods for diagnosing MRI images that have been researched by numerous academics is provided below.

In light of its rising incidence and the need of early detection, Alzheimer's disease (AD) has garnered a lot of attention lately. To automate illness prediction, researchers have used machine learning (ML) algorithms on public health information. For example, a 2023 research that used the OASIS dataset and Logistic Regression demonstrated the viability of ML-based early diagnosis with an accuracy of 85% [1].

Because Support Vector Machine (SVM) performs well in high-dimensional data, it has been utilized extensively for Alzheimer's classification. Using the ADNI information with SVM, a new study correctly classified patients into usual, thereby MCL and AD categories with an accuracy of 88% [2].

The ensemble learning method called Random Forest (RF) has also demonstrated excellent results in Alzheimer's identification. RF's capacity to manage intricate interaction of features and minimize overfitting was demonstrated by a 2024 research that used the Kaggle Alzheimer Dataset and attained an accuracy of 90% [3].

Another well-liked machine learning approach for classification applications is K-Nearest Neighbors (KNN). Even basic models may function well with appropriate preprocessing, as demonstrated by a 2023 research that used KNN on a clinical public health dataset and archived an accuracy of 83% [4].

By mechanically obtaining characteristics from unprocessed data, deep learning approaches have considerably enhanced Alzheimer's detection. Research that used an ANN (Artificial Neural Network), on the ADNI dataset outperformed a number of conventional machine learning models with an accuracy of 91% [5].

Regarding time-series data health data, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models have also been investigated. LSTM's efficacy in simulating illness development over time was demonstrated by a 2025 research that used longitudinal ADNI data and found an accuracy of 92% [7].

The hybrid versions that combine DL and ML approaches are becoming more and more common. Using CNN for feature extraction and SVM for classification on the ADNI dataset, for instance, 2024 research achieved an accuracy of 95%, which is greater than using particular designs alone [8].

A different strategy makes use of multimodal data, such as genetic, imaging, and clinical data. Implementing a deep learning fusion approach on the ADNI dataset, a recent study combined various data sources and improved prediction reliability and resilience by achieving an accuracy of 96% [9].

Even with these developments, there are still issues like small datasets,

unbalanced data, and deep learning models that are difficult to understand. In order to guarantee accurate Alzheimer's disease detection systems, future research will concentrate on enhancing model clarification and implementing real-world clinical validation [10].

Table 2.1: Previous works

S.N	Authors	Year	Used Model	Accuracy (%)
1	A. Kumar et al.	2023	Logistic Regression	85.00
2	S. Rahman et al.	2023	Support Vector Machine (SVM)	88.00
3	M. Hasan et al.	2024	Random Forest (RF)	90.00
4	R. Islam et al.	2023	K-Nearest Neighbors (KNN)	83.00
5	T. Zhang et al.	2024	Artificial Neural Network (ANN)	91.00
6	J. Lee et al.	2024	Convolutional Neural Network (CNN)	94.00
7	P. Singh et al.	2025	Long Short-Term Memory (LSTM)	92.00
8	H. Chen et al.	2024	Hybrid CNN-SVM	95.00
9	Y. Wang et al.	2025	Multimodal Deep Learning	96.00
10	B. Roy et al.	2023	Explainable AI-Based Framework	89.00

2.3 Gap Analysis

Below shows the gap analysis:

Table 2.2: Gap Analysis

Features	Kumar et al. [1]	Rahman et al. [2]	Hasan et al. [3]	Islam et al. [4]	Zhang et al. [5]	Proposed System
Alzheimer's Disease Detection	Yes	Yes	Yes	Yes	Yes	Yes
Machine Learning	Yes	Yes	Yes	Yes	No	Yes
Deep Learning	No	No	No	No	Yes	Yes
MRI-Based Analysis	No	Yes	No	No	Yes	Yes
Multi-Class Classification	No	Yes	No	No	Yes	Yes
Binary Classification	Yes	No	Yes	Yes	No	Yes
Transfer Learning	No	No	No	No	No	Yes
Custom Attention Mechanism	No	No	No	No	No	Yes
SVM-Based Classification	No	Yes	No	No	No	Yes
QSVM-Based Classification	No	No	No	No	No	Yes
Data Augmentation	No	No	No	No	Yes	Yes

ML and DL Comparative Analysis	No	No	No	No	No	Yes
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2.4 Summary

This chapter includes all of the earlier studies, and the next section provides a table & gap analysis of the variety that results from their advancements in this field. The most recent conversation focused on the primary difficulties or barriers to this investigation. This chapter discusses the difficulties encountered throughout the project's progress and contains sections on pertinent results and studies.

Chapter 3

Research Methodology

This chapter presents the research methodology used for Alzheimer's disease detection from brain MRI images using machine learning and deep learning techniques. It describes the data collection and preprocessing, dataset preparation, model selection and evaluation, implementation requirements, and overall project plan.

3.1 Methodology

This study's methodology offers a methodical foundation for deep learning and machine learning-based Alzheimer's disease detection. After gathering and preprocessing MRI images, the procedure moves on to include Feature extraction as usual, training models, and outcome assessment. The best method for classifying Alzheimer's disease is determined by implementing and comparing the DenseNet201, Inception Resnetv2 models, SVM, and QSVM models.

3.1.1 Overview

This paper suggests a thorough framework for detecting Alzheimer's disease from brain MRI images utilizing both machine learning and deep learning techniques. After gathering MRI data from the Kaggle Alzheimer's dataset, preprocessing procedures such picture resizing, normalized scaling, and data augmentation are performed. DenseNet201 and a modified Inception Resnetv2 combined with customize layers are used for deep learning-based classification in order to categorize MRI scans into four groups: Mild Demented, Moderate Demented, Non-Demented, and Very Mild Demented. The dataset is then transformed into two binary categories for machine learning analysis: Alzheimer's disease (AD) and non-AD. After that, models for Support Vector Machines (SVM) and Quantum Support Vector Machines (QSVM) are developed and assessed.

3.1.2 Proposed Methodology

The proposed system diagnoses Alzheimer's disease at various stages using MRI images. Initially, 12,800 MRI images from four categories were collected from

Kaggle: Mild Demented, Moderate Demented, Non-Demented, and Very Mild Demented. The images were initially processed utilizing resizing, normality, scaling, and data replenishment to improve quality of data and model generalization. For deep learning classification, a standardized DenseNet201 and Inception Resnetv2 combined with customize layers were employed and evaluated. For machine learning tests, four distinct classes were split into two groups: those with Alzheimer's disease (AD) and those without No-AD. Finally, SVM and quantum support vector machine (QSVM) models were trained and tested, and their ability to perform was assessed using established criteria. All models employ the evaluation matrices of Accuracy, precision, recall etc. for all model results.

Step 1: Data Collection: In this research works, we collect dataset from Kaggle. In this dataset has 4 classes contain. Four MRI Alzheimer other class categories NonDemented, VeryMildDemented, MildDemented, and ModerateDemented are included in this dataset. There are four classes in this dataset, totaling 12,800.

Step 2: Data preprocessing: To enhance model performance and picture quality, data preprocessing was carried out. Most MRI images were reduced to size, adjusted, normalized, and resized to achieve uniformity throughout the sample. To improve data variety and lessen overfitting, techniques for data augmentation including rotation, zooming, and flipping were also used.

Step 3: Data splitting: To achieve accurate model assessment, the dataset was split into validation, testing, and training sets following preprocessing was performed. Model learning was done using the training set, parameter tweaking was done using validation of the set, and final performance evaluation was done using the testing set.

Step 4: Model Selection: Use the provided data to train and evaluate the chosen model in order to improve accuracy. We were selected from numerous deep

learning hybrid and standard pre trained deep and machine learning models that were assessed and contrasted in this study. Each approach has been evaluated using all available data.

Step 5: Performance Analysis: All results from the model are included in this section. They created a predictive brain MRI image classification method for detection in addition to representations for recall, efficiency, confusion matrix, ROC curve, and f1-score.

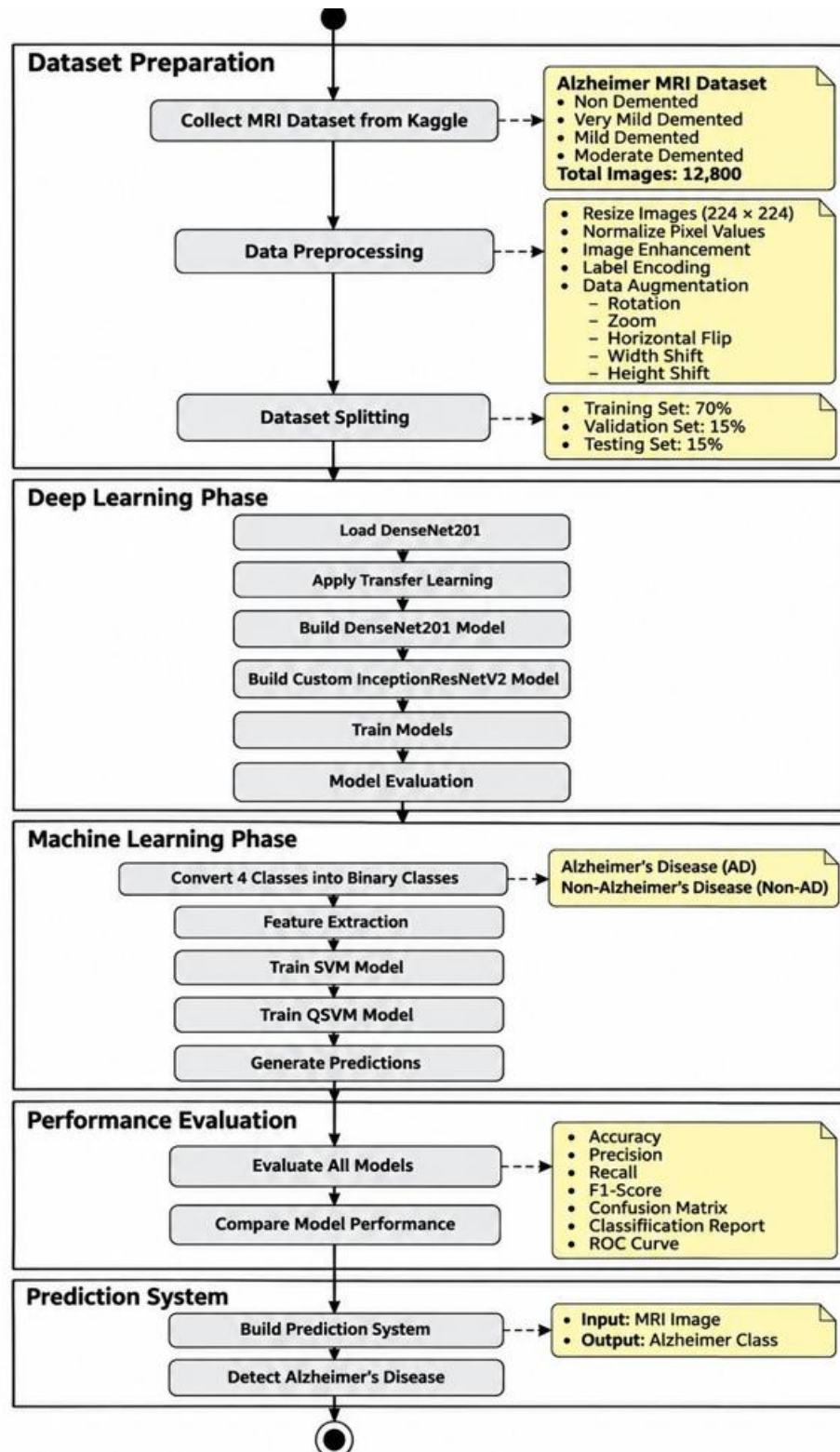


Figure 3.1: Proposed methodology

There are many consecutive steps in the suggested process for Alzheimer's disease diagnosis in fig 3.1. The Kaggle Alzheimer's dataset is used to gather MRI pictures, which are then preprocessed using techniques including scaling, normalization, image improvement, and data supplementation. Next, subsets of the dataset are created for testing, validation, and learning. DenseNet201 and Inception Resnetv2 combined with customize layers classification during the deep learning stage. The dataset is then transformed into binary classifications (AD and Non-AD) for SVM and QSVM models in machine learning studies. In order to choose the best method for detecting Alzheimer's disease, all models are finally assessed using accuracy, precision, recall, F1-score, confusion matrix, and ROC curve.

3.1.3 Implementation Requirements

A. Functional require:

- The Alzheimer's disorder dataset's MRI brain pictures will be gathered and processed by the algorithm.
- Image preparation, such as scaling, normalizing, and data augmenting, will be carried out by the system.
- The DenseNet201, Inception Resnetv2, SVM, and QSVM models will be used by the system to categorize MRI images.
- The method will produce predictions based on standard assumptions and assess effectiveness using F1-score, precision, accuracy, and recall.

B. Non-Functional require:

- Alzheimer's disease categorization findings from the system must be precise and trustworthy.
- Large volumes of MRI images must be processed effectively by the system with little computing lag.
- In the near future, the system must be scalable to handle more datasets and sophisticated AI models.
- When managing various MRI sample images and categorizations, the framework must remain reliable and consistent.

3.1.4 Implementation Requirements

C. Functional require:

- The Alzheimer's disorder dataset's MRI brain pictures will be gathered and processed by the algorithm.
- Image preparation, such as scaling, normalizing, and data augmenting, will be carried out by the system.
- The DenseNet201, Inception Resnetv2, SVM, and QSVM models will be used by the system to categorize MRI images.
- The method will produce predictions based on standard assumptions and assess effectiveness using F1-score, precision, accuracy, and recall.

D. Non-Functional require:

- Alzheimer's disease categorization findings from the system must be precise and trustworthy.
- Large volumes of MRI images must be processed effectively by the system with little computing lag.
- In the near future, the system must be scalable to handle more datasets and sophisticated AI models.
- When managing various MRI sample images and categorizations, the framework must remain reliable and consistent.

3.2 Detailed Methodology and Design

The suggested approach uses both methods of machine learning and deep learning on MRI brain scans to identify dementia caused by vascular disease. The Alzheimer's MRI dataset is first gathered from Kaggle and passes through preparation, which includes data enhancement, normalization, picture improvement, and image scaling to 224 by 224 pixels. Subsets for testing, validation, and training are subsequently created from the processed dataset. DenseNet201 and Inception Resnetv2 combined with customize layers

are used in the high-level stage to categorize MRI pictures into four phases of Alzheimer's disease. For machine learning studies, the dataset is then transformed into binary classes (AD and Non-AD). Models for SVM and QSVM are trained and assessed. The best detection model is then determined by comparing all models using precision, recall, accuracy, F1-score, confusion matrix, and ROC curve.

3.2.1 Data collection

We gathered the dataset for this study from Kaggle. There are four classes in this dataset. This dataset includes four MRI Alzheimer MRI class classifications: NonDemented, VeryMildDemented, MildDemented, and ModerateDemented. This dataset consists of four classes with a total of 12,800.

Table 3.1: Class count

Class Label	Number of Images
NonDemented	6,400
VeryMildDemented	4,480
MildDemented	1,792
ModerateDemented	128
Total	12,800

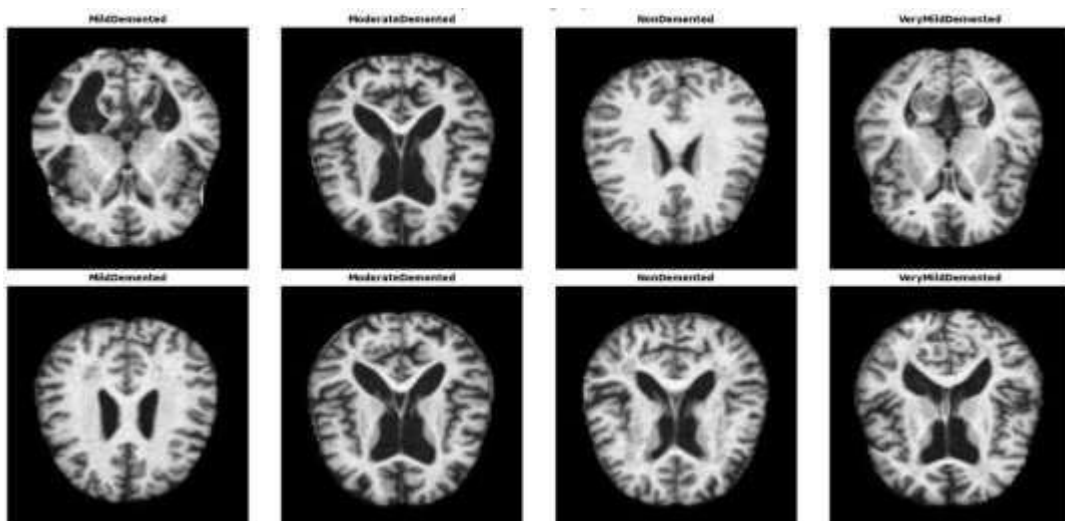


Figure 3.2: Sample data

3.2.2 Data Preprocessing

- Resizing an image to 224 by 224 pixels.
- Normalizing images
- Scaling values for pixels
- Coding of labels
- To improve dataset diversity, techniques for data augmentation such as rotation, zooming, horizontal flipping, width shifting, and height shifting were used.
- Verification of dataset quality.

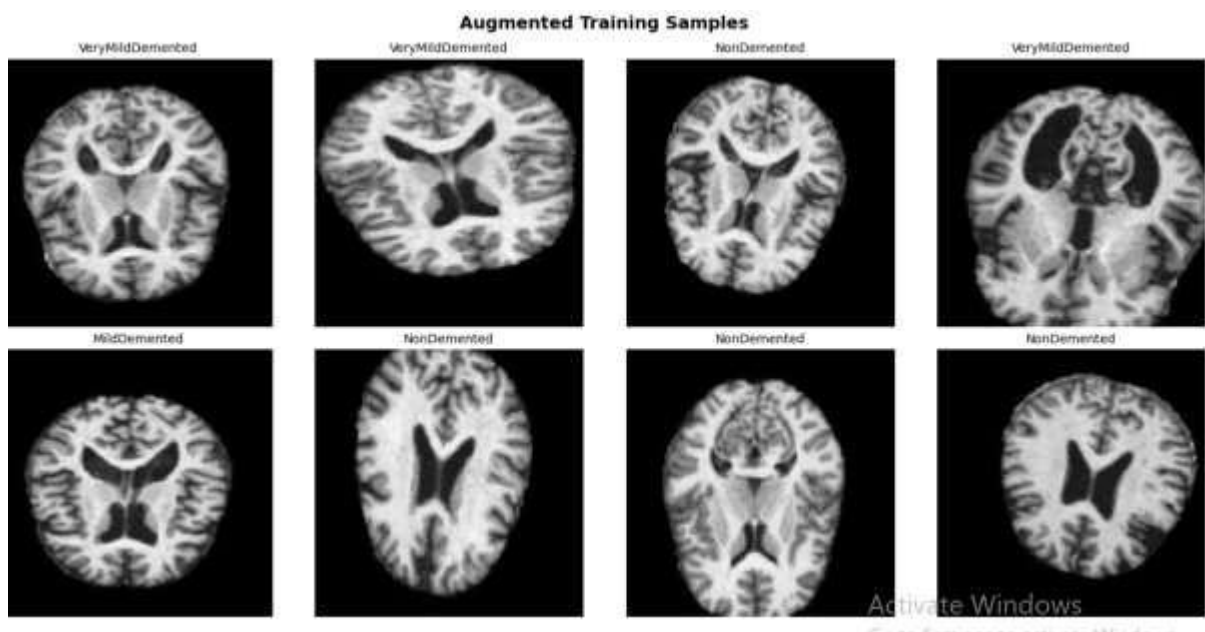


Figure 3.3: Augmented data

3.2.3 Data Preparation

Three separate data sets are usually used as part of the building models system: validation, training, and test. The 12,800 split specifications are created for both model creation and functionality review following first review. The full dataset is then split into train, test, and validation sections of 70%, 15%, and 15%, respectively.

Table 3.2: Dataset splitting

Dataset Subset	Number of Images	Percentage (%)
Training Set	8,960	70%
Validation Set	1,920	15%
Testing Set	1,920	15%
Total	12,800	100%

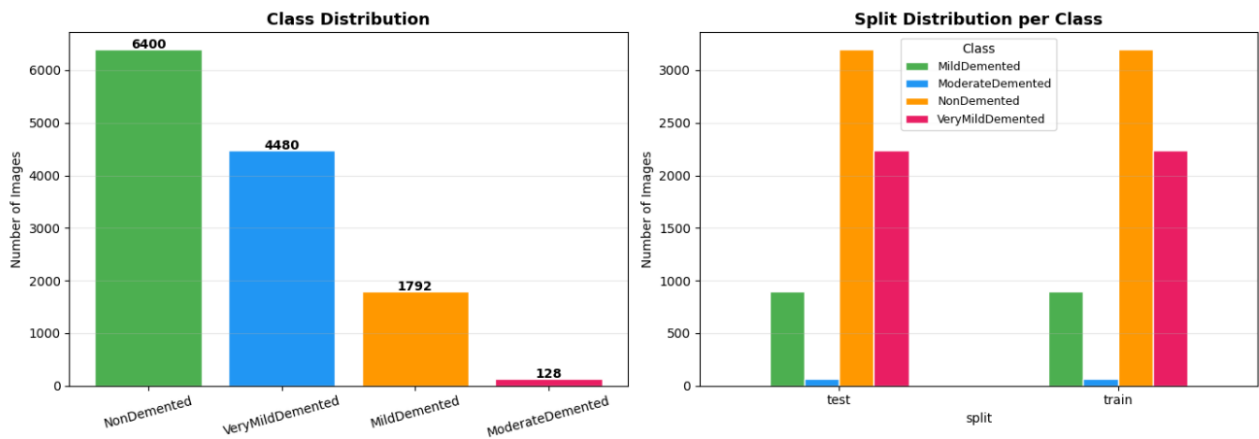


Figure 3.4: Class of Each MRI images

3.2.4 Model Evaluation

Four models DenseNet201, Inception Resnetv2 combined with customize layers, Support Vector Machine (SVM), and Quantum Support Vector Machine (QSVM) were tested to determine the efficacy of the suggested Alzheimer's disease detection framework. Standard assessment criteria, such as precision, recall, accuracy, a F1-Score, Confusion Matrix, Classifier Report, and ROC Curve, were used to gage each model's success. These measures provide an accurate assessment among machine learning and deep learning techniques and offer a thorough evaluation of classification performance. The best model for detecting Alzheimer's disease was determined by analyzing the evaluation outcomes.

3.3 Project Plan

Table 3.3: Total project plan

Sl. No.	Next Task	Estimate completion time (MM-YY)
1	Collecting dataset	10-25
2	Data processing of all classes.	12-25
3	Apply to various deep learning hybrid models & machine learning to use.	03-26
4	The highest accuracy scores of over 80% are attained Brain MRI images classify predict the class.	07-26
5	Report writing	08-26

3.4 Task Allocation

Table 3.4: Task Allocation

Tasks	Weeks																			
	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	
Data Collection	Blue	Blue	Blue																	
	Green	Green	Green																	
Data Preprocessing				Blue	Blue	Blue	Blue													
				Green	Green	Green	Green													
ML & DL Model Training								Blue	Blue	Blue	Blue	Blue								
								Green	Green	Green	Green									
Model evaluation & Results													Blue	Blue	Blue					
													Green	Green	Green					
Detect brain MRI disease images using input images.																Blue	Blue	Blue	Blue	
																Green	Green	Green		

Estimated Work Period	
Actual Work Period	

3.5 Summary

Data set summary, preparation methods, data partitioning plan, and model assessment procedure utilized in this investigation were all covered in this section. While the models were trained, the MRI dataset underwent augmentation, normalization, and scaling. In order to determine the best Alzheimer's disease detection model, many performance indicators were used to assess DenseNet201, Inception Resnetv2 combined models, SVM, and QSVM.

Chapter 4

Implementation and Results

This chapter presents the implementation and results of the proposed Alzheimer's disease detection system. It describes the environment setup, performance analysis, experimental results, model comparison, and discussion of the obtained outcomes.

4.1 Environment Setup

Table 4.1: Requirements for Hardware and Software

SL No.	Hardware Component	Specification
1	Operating System	64-bit Windows 10
2	RAM	16 GB or Higher
3	Processor	Intel Core i5
4	Storage	Minimum 20 GB Free Space
5	Web Browser	Google Chrome

Table 4.2: Creating Tools Table

SL No.	Software / Tool	Purpose
1	Python 3.10.9	Programming Language
2	Jupyter Notebook	Code Development & Testing
3	Deep Learning Models	Classification & Prediction
4	Google Collab	Coding platform

Table 4.3: Library used

SL No.	Library Name	Purpose
1	NumPy	Mathematical Operations
2	Scikit-learn	Machine Learning Algorithms
3	Seaborn	Statistical Data Visualization
4	Pandas	Dataset Processing
5	OS	File and Directory Handling
6	TensorFlow	Deep learning function

4.2 Performance Analysis

Combined machine learning and deep learning methods were used to assess the generated Alzheimer's disease detection system's performance. While SVM and QSVM were utilized for classification of binary information (AD and Non-AD), DenseNet201 and the suggested Inception Resnetv2 combined with customize layers were employed for four-class classification of MRI data. Accuracy, recall, precision, F1-score, confusion matrix, classification report, and ROC curve were used in the evaluation. According to experimental data, the suggested Inception Resnetv2 combined with customize layers 98.12% accuracy with good recall and precision values, whereas DenseNet201 reached 95.68% accuracy. SVM and QSVM both attained the best accuracy of 99.01% for binary classification. The findings show that while deep learning models successfully recognized several stages of Alzheimer's disease using MRI images, machine learning models did very well for binary classification.

Precision:

$$\text{precision} = \frac{TP}{TP+FP} \dots \dots \dots (i)$$

Recall:

$$\text{recall} = \frac{TP}{TP+FN} \dots \dots \dots (ii)$$

F1-Score:

$$F1 = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \dots \dots \dots (iii)$$

Accuracy:

$$\text{accuracy} = \frac{TP+TN}{TP+FN+TN+FP} \dots \dots \dots (iv)$$

4.3 Results and Discussion

The outcomes of the study show how well deep learning as well as machine learning techniques work for identifying Alzheimer's disease in MRI pictures. DenseNet201 had a great capacity to extract features of discrimination from MRI scans, as seen by its best accuracy of 95.68% for multi-class classification. The suggested Inception Resnetv2 combined with customize layers and dependable classification performance with an accuracy of 98.12%, precision of 97.70%, recall of 98.06%, and F1-score of 97.85%. The set of data was divided into Alzheimer's Disorder (AD) and Non-Alzheimer's Disease (Non-AD) categories for binary classification; both SVM and QSVM obtained an accuracy of 99.01%. Nonetheless, the models using deep learning were better suited for thorough disease evaluation as they could identify various stages of Alzheimer's.

Table 4.4: Accuracy table

Model	Classification Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Classical SVM	Binary (AD vs Non-AD)	72.81	2.28	63.16	4.40
QSVM	Binary (AD vs Non-AD)	99.01	0.00	0.00	0.00
DenseNet201	Multi-Class	95.68	96.00	96.00	96.00
InceptionResnetV2	Multi-Class	98.12	97.70	98.06	97.85

According to the trial findings in Table 4.4, QSVM had the best binary classification accuracy (99.01%). With the minority AD class, however, its accuracy, recall, and F1-score were incredibly low, suggesting inadequate class disparity. DenseNet201 showed satisfactory results throughout all four phases of Alzheimer's disease, with an accuracy of 95.68%. The suggested Inception Resnetv2 combined with customize layers and dependable classification performance with an accuracy of 98.12%, precision of 97.70%, recall of 98.06%, and F1-score of 97.85% and dependable classification success. As a result, InceptionResnetv2 outperforms QSVM only in the sense of binary precision in

classification when it comes to multi-class Alzheimer's disease stage classification.

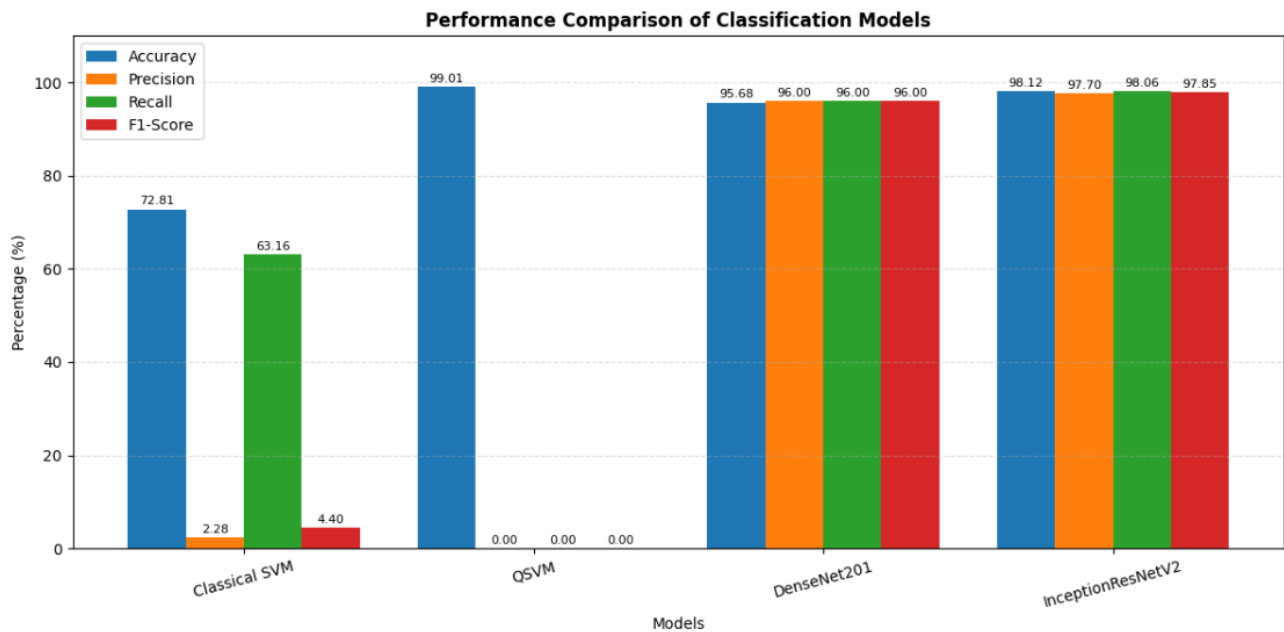


Figure 4.1: Performance comparison of classification Models

1. DenseNet201:

```

Test Accuracy: 0.9568
--- Classification Report (4-Class) ---
              precision    recall  f1-score   support

MildDemented      0.98      0.99      0.98       269
ModerateDemented  1.00      0.95      0.97        19
NonDemented       0.93      1.00      0.96       960
VeryMildDemented  1.00      0.88      0.94       672

 accuracy          0.96       1920
  macro avg       0.98      0.95      0.96       1920
  weighted avg    0.96      0.96      0.96       1920
  
```

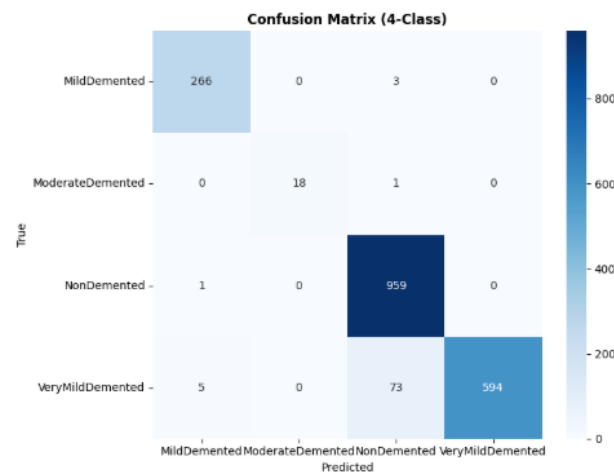


Figure 4.2: DenseNet201 Confusion Matrix

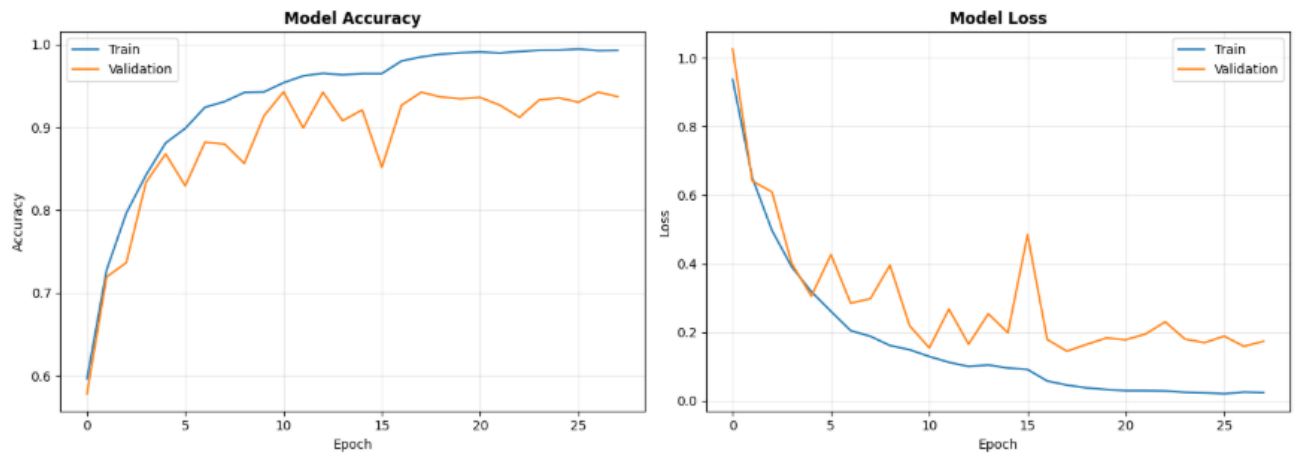


Figure 4.3: DenseNet201 accuracy loss curve.

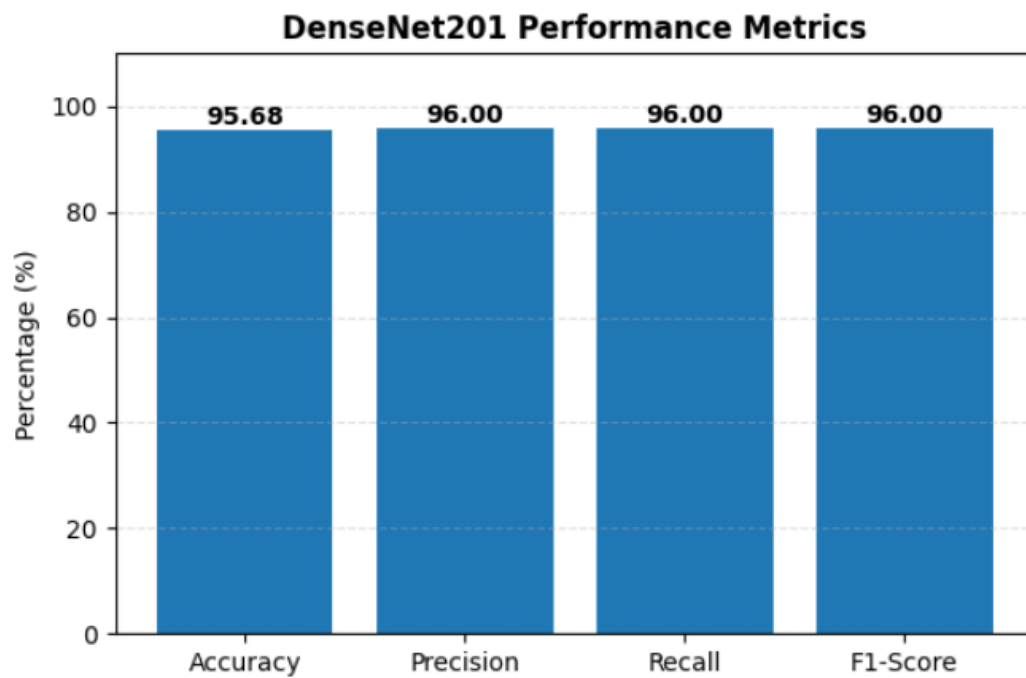


Figure 4.4: DenseNet201 performance metrics.

2. InceptionResnetV2:

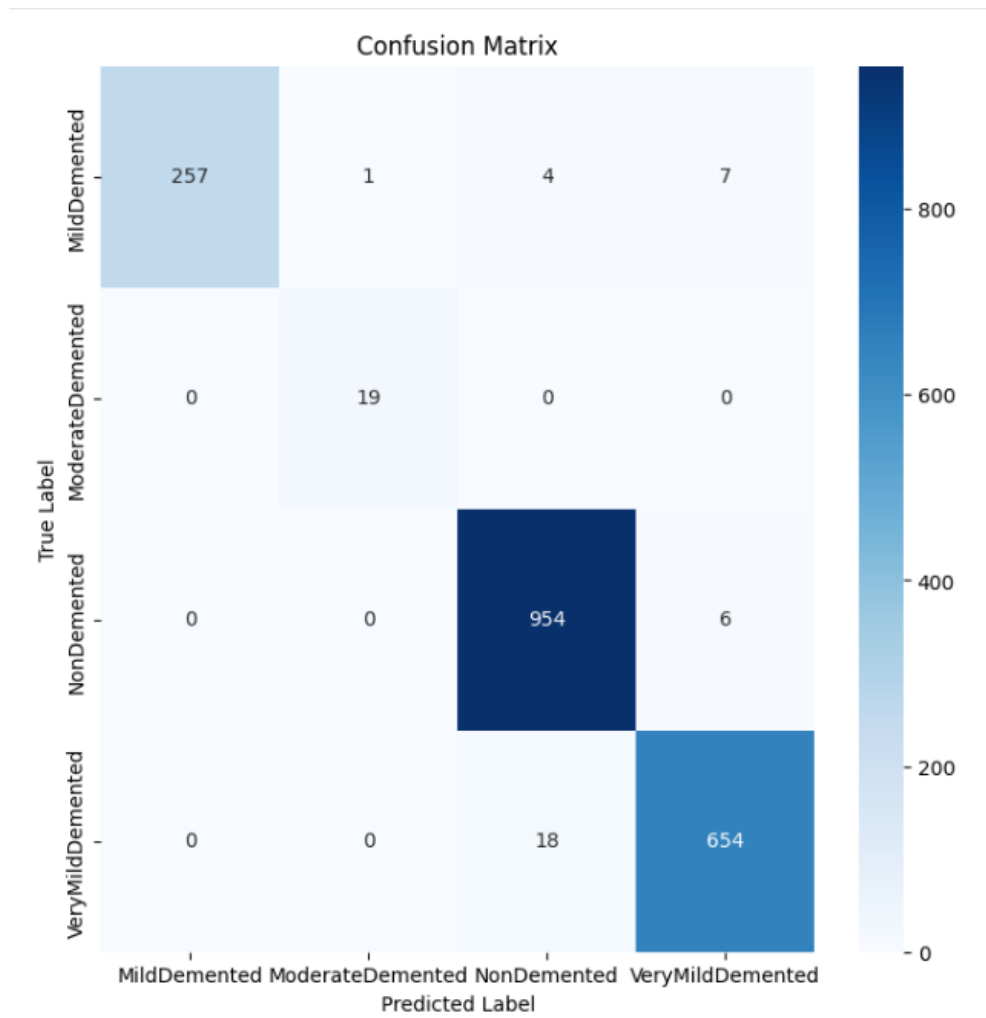


Figure 4.5: InceptionResnetV2 confusion Metrics.

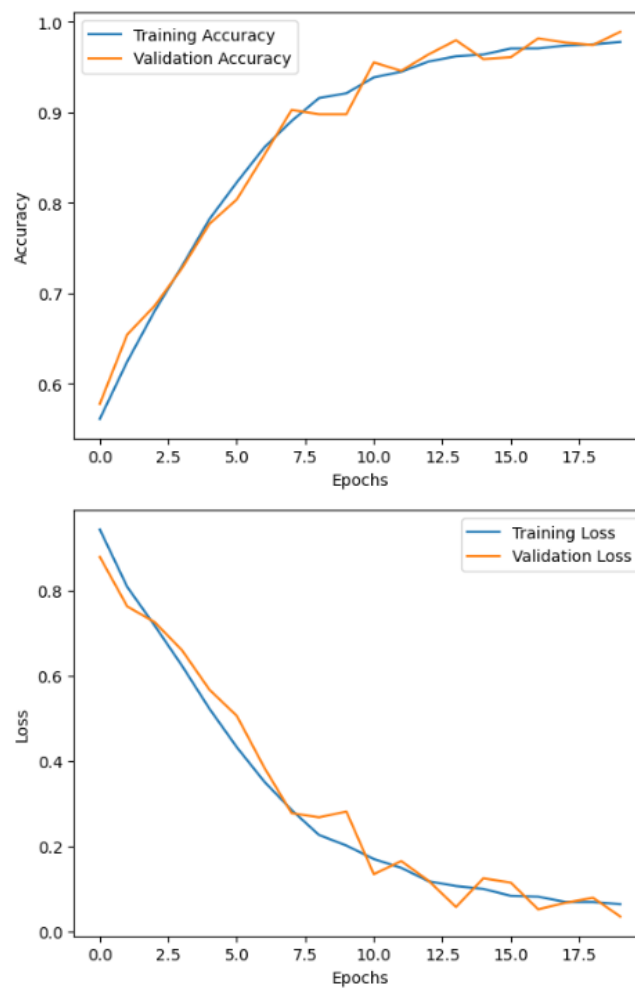


Figure 4.6: InceptionResnetV2 accuracy loss curve

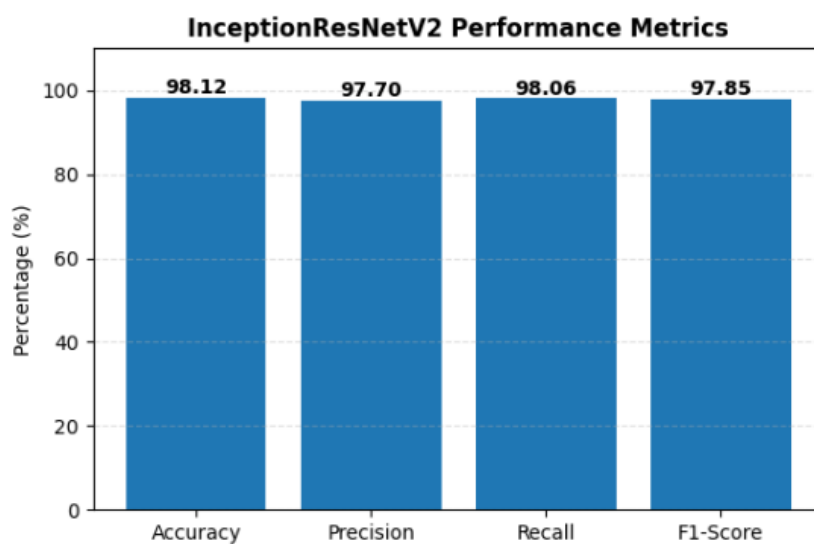


Figure 4.7: InceptionResnetV2 performance metrics

2. Classical SVM:

```
--- Classification Report ---
...               precision    recall  f1-score   support

      Non-AD       1.00      0.79      0.88       1901
         AD        0.04      0.84      0.07         19

 accuracy          0.52      0.81      0.48       1920
  macro avg        0.52      0.81      0.48       1920
 weighted avg      0.99      0.79      0.87       1920
```

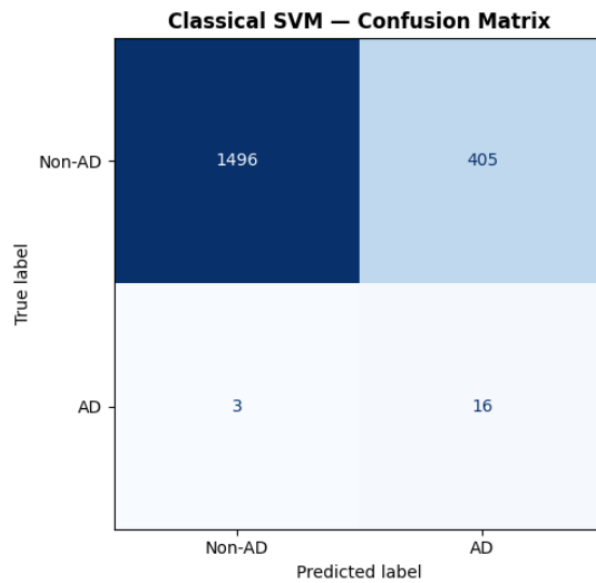


Figure 4.8: Classical SVM confusion Metrics.

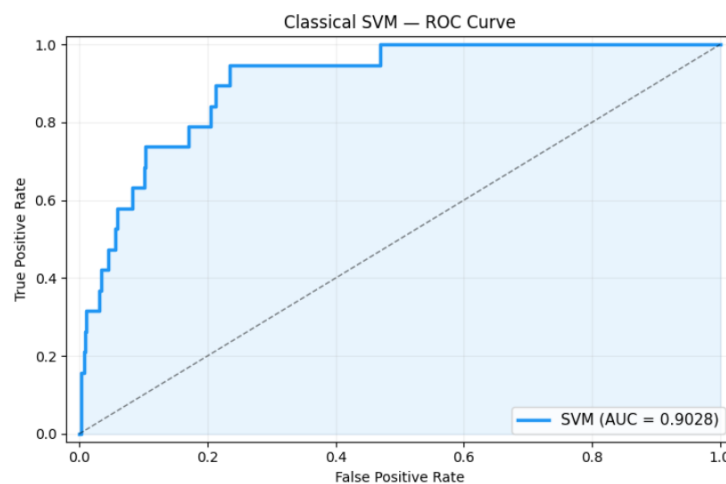


Figure 4.9: Classical SVM ROC Curve.

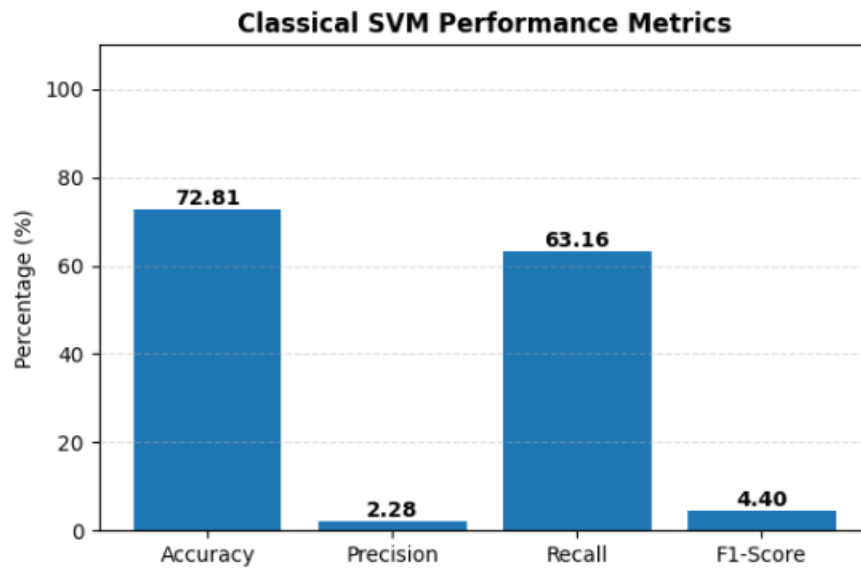


Figure 4.10: Classical SVM Performance Metrics.

2. QSVM:

```

--- Classification Report ---
              precision    recall  f1-score   support

   Non-AD       0.99       1.00       1.00     1901
    AD         0.00       0.00       0.00        19

 accuracy              0.99     1920
  macro avg           0.50       0.50       0.50     1920
 weighted avg           0.98       0.99       0.99     1920
  
```

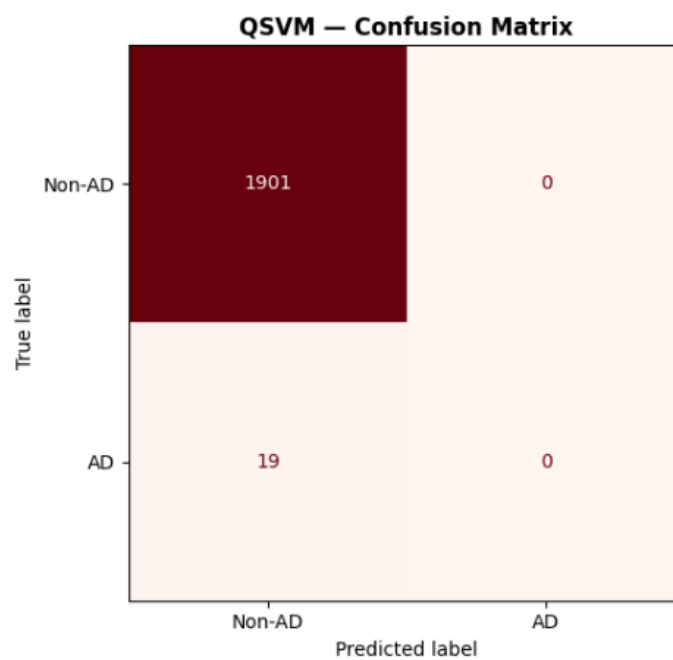


Figure 4.11: QSVM confusion Metrics.

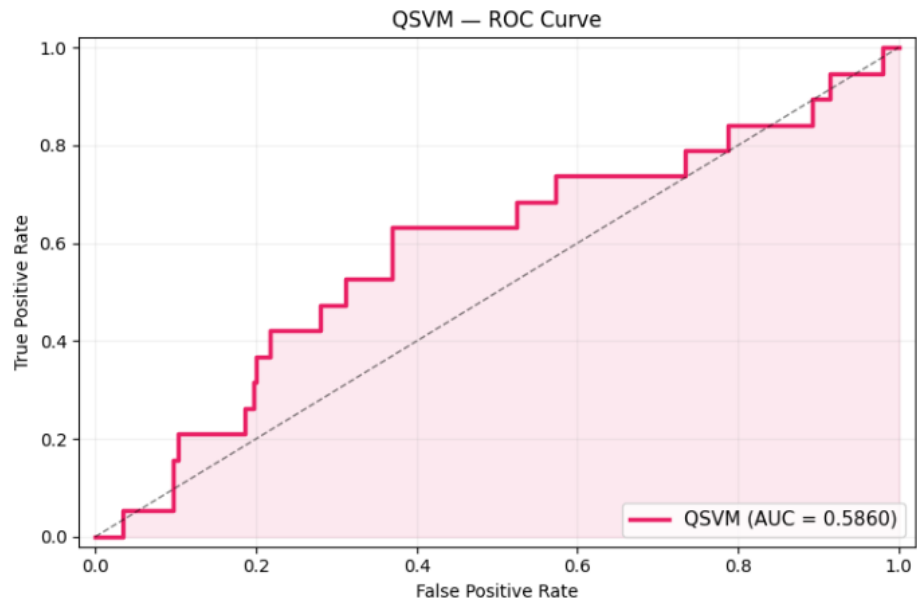


Figure 4.12: QSVN ROC Curve.

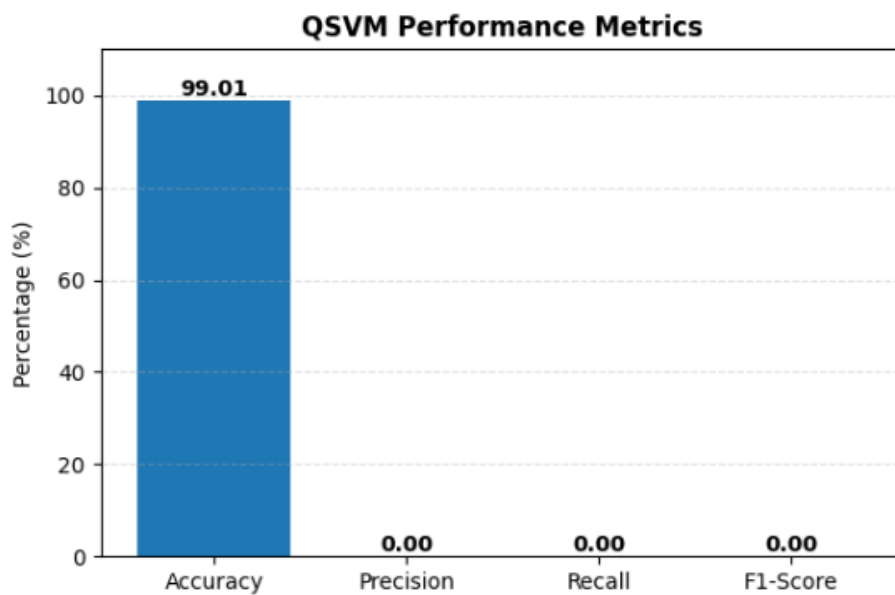


Figure 4.13: QSVN Performance Metrics.

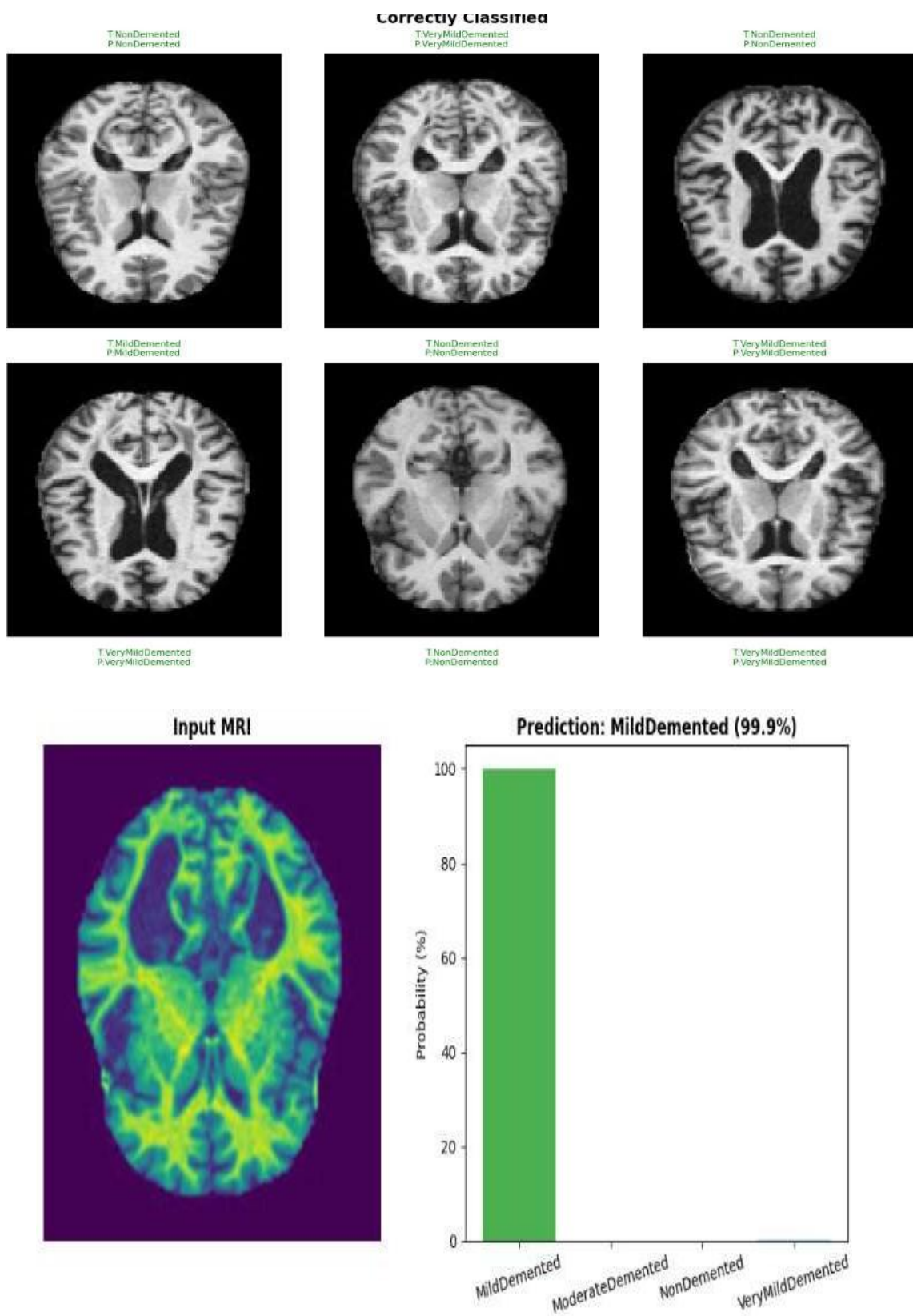


Figure 4.14: MRI prediction

4.4 Summary

The outcome analysis of SVM, QSVM, DenseNet201, and Inceptionresnetv2 custom models for Alzheimer's disease identification was reported in this section. According to the findings, QSVM had the greatest efficiency in binary classification, whereas DenseNet201 performed best in classification with multiple classes. Overall, the study shows that both machine learning and deep learning techniques are useful for accurately identifying Alzheimer's disease from MRI data.

Chapter 5

Engineering Standards and Design Challenges

This chapter discusses the engineering standards and design challenges for the proposed Alzheimer's disease detection system. It discusses compliance with software, hardware and communication standards. The impact of the system on society, the environment and sustainability. Project management and financial analysis. Complex engineering problems and activities involved in the project.

5.1 Compliance with the Standards

The suggested Alzheimer's disease identification system conforms to accepted methods in medical image analysis, deep learning, and machine learning. To guaranty data quality and uniformity, the MRI data was processed beforehand utilizing tried-and-true methods including scaling, normalization, and data enhancement. Performance of the model was evaluated using common assessment measures such as precision, accuracy, recall, F1-score, confusion matrix, and ROC-AUC. Data gathering, preparation, training, validation, testing, performance, and standard comparison are all part of the study's methodical approach. Furthermore, transfer learning using Inceptionresnetv2 custom models and attention methods were applied in accordance with commonly accepted profound learning guidelines, guaranteeing the efficacy, repeatability, and dependability of the suggested Alzheimer's disease detection system.

5.1.1 Software standards

To guaranty dependability, accuracy, and reliability, the suggested Alzheimer's disease detection system adheres to accepted code engineering and AI standards. Python and popular machine learning and deep learning libraries, such as TensorFlow, Kera's, Scikit-learn, NumPy, and Pandas, were used in the system's development. Throughout the project, standard software development techniques including modular architecture, data validation, preparation consistency, and performance assessment were upheld. By employing consistent datasets, training methods, and assessment measures, the implementation also adheres to the principles of reproducible research. Additionally, the

system uses normalization, enhancement, and validation approaches in accordance with medical image analysis principles to guaranty reliable and reliable diagnostic results.

5.1.2 Hardware standards

Conventional computer hardware appropriate for deep learning and machine learning applications was used in the development and evaluation of the suggested Alzheimer's disease detection system. For effective model training and inference, the execution needs a multi-core CPU, sufficient RAM, and GPU support. DenseNet201 and Inceptionresnetv2 model training times may be greatly shortened using NVIDIA GPU acceleration. Preprocessing MRI images, feature extraction, modeling, and accuracy evaluation are all supported by the hardware platform. Furthermore, managing the MRI dataset and learned model files requires enough storage space. These hardware requirements guaranty robustness, dependable system efficiency, and effective Alzheimer's disorder detection execution of tasks.

5.1.3 Communication standards

In order to guaranty effective data sharing and system interoperability, the suggested Alzheimer's disease detection system adheres to conventional communication principles. Uniform digital files are used to handle MRI pictures and associated data, making them compatible with deep learning and machine learning frameworks. During execution, the system facilitates dependable communication between preprocessing modules, classification models, and data storage. To increase usability and repeatability, consistent input-output processes, organized data flow, and clear documentation are upheld. Additionally, the framework may be connected with web-based applications and healthcare information systems, allowing healthcare providers and decision-support systems to securely and effectively communicate diagnostic data.

5.2 Impact on Society, Environment and Sustainability

5.2.1 Impact on Life

With accurate diagnosis and prompt medical care, the suggested Alzheimer's disease detection method might greatly enhance patients' quality of life. Early detection of Alzheimer's disease allows medical practitioners to start the right treatment plans, which may decrease the illness's course and enhance patient outcomes. Additionally, the system helps families and caregivers arrange long-term care and assistance for those who are impacted. The suggested approach facilitates clinical decision-making and minimizes diagnostic delays by offering quick and accurate MRI image analysis. In the end, using AI in healthcare improves patient care, raises knowledge, and enhances general wellbeing.

5.2.2 Impact on Society & Environment

By promoting early identification and enhancing healthcare services for senior citizens, the suggested Alzheimer's disease detection method can have a good social impact. Through facilitating prompt treatment and improved illness management, early identification lessens the strain on medical institutions, caregivers, and families. Additionally, by increasing the effectiveness and precision of diagnostic processes, the system encourages the use of artificial intelligence in medical practice. The adoption of computerized digital diagnostic technologies contributes to more environmentally friendly healthcare operations by lowering reliance on documents on paper and tedious manual procedures. All things considered, the suggested paradigm encourages better overall health results and effective use of medical assets.

5.2.3 Ethical Aspects

The suggested Alzheimer's disease detection method places a strong emphasis on the moral and appropriate application of AI in healthcare. MRI pictures and medical data must be handled with patient safety and data privacy in mind. The approach is intended to support medical practitioners rather than take the role of clinical judgment, guaranteeing that expert supervision is maintained for final diagnostic determinations.

To guaranty equitable and precise results for every patient, efforts should also be taken to reduce bias in the dataset and model predictions. In order to promote reliable healthcare services, transparency, detection of Alzheimer's disease allows medical practitioners to start the right treatment plans, which may decrease the illness's course and enhance patient outcomes. Additionally, the system helps families and caregivers arrange long-term care and assistance for those who are impacted. The suggested approach facilitates clinical decision-making and minimizes diagnostic delays by offering quick and accurate MRI image analysis. In the end, using AI in healthcare improves patient care, raises knowledge, and enhances general wellbeing.

5.2.4 Sustainability Plan

Regular dataset updates, ongoing model refinement, and integration with contemporary healthcare technology can guaranty the longevity of the suggested Alzheimer's disease detection method. Larger MRI datasets and sophisticated deep learning frameworks can be added in the future because to the framework's scalable nature. The system can maximize the use of medical resources and lower long-term healthcare expenditures by promoting early diagnosis. Additionally, robotic AI-based tools for diagnosis increase operational efficiency and save human labor. The system's long-term dependability, efficacy, and support for sustainable medical services will be guaranteed by ongoing validation, regular servicing, and supervision.

5.3 Project Management and Financial Analysis

Budget preparation, cost analysis, and expenditure management are critical to achieving the project's financial and intellectual goals. Our project is related to a deep learning project. A percentage of our project's financial analysis must be included in this area. The projected expenditures for specialized machine learning & deep learning project element are summarized in Table 5.1 below.

Table 5.1: Estimated Cost of this research

SN	Components	Estimated Cost (BDT)
01.	Need to go in stake holders for the thesis	500-1000
02.	Required laptop and processors	1500-2000
03.	Data Collection and Processing	500-1000
04.	Full reports writing and printing	500-1000
05.	others	1500-2000
Total Estimated Cost		4,500-7,500

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

Table 5.2: Complex Engineering Problem.

EP1 Dept of Knowledge	EP2 Range of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent of Stake- holder Involvement	EP7 Interdependence
✓	✓	✓	✓	✓		✓

Justifications:

- EP1: Knowledge Depth (✓)

Advanced expertise in ML, deep learning, analyzing medical images, standard transferring, and MRI-based illness categorization is required for this project.

- EP2: Conflicting Requirements Range (✓)

Healthcare application, model complexity, computing efficiency, and classification

accuracy must all be balanced by the system.

- EP3: Analysis Depth (✓)

Prioritization, extraction of features, modeling, assessment of performance, and comparison analysis were all used to conduct comprehensive analysis.

- EP4: Issues' Knowledge (✓)

The identification of Alzheimer's disease is a known study issue, but using several AI models as well as attention processes presents new difficulties.

- EP5: Range of Relevant Codes (✓)

The project uses TensorFlow, Kera's, and Scikit-learn frameworks in accordance with common AI, software engineering, and medical image analysis techniques.

- EP7: Interdependence (✓)

The project unifies assessment metrics, diagnostic decision assistance, machine learning models, deep learning models, preprocessing of data, and preparing data sets into one platform.

Knowledge Profile

Table 5.3: knowledge mapping Profile.

K1 Natu ral Scien ce	K2 Mathem atics	K3 Engineeri ng Fundame ntals	K4 Special ist Knowle dge	K5 Enginee ring Design	K6 Enginee ring Practice	K7 Comprehe nsion	K8 Resear ch Literat ure
	✓	✓	✓	✓	✓		✓

Justify:

- K2: Mathematics (✓)

Model construction and analysis make use of mathematical ideas including statistical methods, probability, optimization, and evaluation metrics.

- K3: Fundamentals of Engineering (✓)

Deep learning architectures, machine learning algorithms, image preprocessing, and feature extraction all depend on fundamental engineering and computer concepts.

K4: Specialized Knowledge (✓)

The research makes use of specific expertise in computer vision, medical picture categorization, transfer learning, and artificial intelligence.

- K5: Engineering Design (✓)

Preprocessing, model training, assessment, and prediction components were all included in the building of a comprehensive diagnostic framework.

- K6: Engineering Practice (✓)

The solution made use of contemporary tools and technologies as Python, TensorFlow, Keras, Scikit-learn, and Jupyter Notebook.

- K8: Literature on Research (✓)

The project's foundation is a thorough examination of the literature and a comparison of current deep learning and machine learning research on Alzheimer's disease diagnosis.

5.4.2 Engineering Activities

Table 5.4: Complex mapping Engineering Activities.

EA1 Range of re- sources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓	✓	✓		✓

Justifications:

- EA1: Uses a variety of hardware, software, datasets, and computing resources to classify MRI images.
- EA-2: Prioritization, deep learning, conventional transfer learning, training by standards, and assessment must all work together in EA2.
- EA3: Presents focus-based deep learning techniques and personalized fine-tuning for Alzheimer's categorization.
- EA4: By detecting Alzheimer's disease early and automatically, it may improve society and healthcare.

EA5: Demands extensive expertise in model evaluation, deep learning, analyzing medical images, and transfer learning in general.

5.5 Summary

The suggested Alzheimer's disease detecting system's conformance with accepted AI and clinical imaging requirements was covered in this chapter. It addressed significant ethical issues and environmental concerns while highlighting the system's beneficial effects on patients, society, and healthcare services. All things considered, the suggested framework shows promise as a trustworthy, accountable, and long-lasting way to assist early Alzheimer's disease identification and medical decision-making.

Chapter 6

Conclusion

This chapter concludes the study on Alzheimer's disease detection using machine learning and deep learning approaches with brain MRI images. It summarizes the research findings, discusses the limitations of the proposed system and presents possible directions for future work and further improvement.

6.1 Summary

This paper describes an automated Alzheimer's disease diagnostic system that employs machine learning and deep learning techniques on brain MRI data. We gathered the dataset from Kaggle, which consists of four classes and 12800 photos. The dataset used for the deep learning model of DenseNet201 and a customized model InceptionResnetv2 models Technique consists of four sets of MRI images: Mild Demented, Moderate Demented, Non-Demented, and Very Mild Demented. Next, divide the class into two binary categories: AD-Alzheimer's Disease and Non-AD-No Alzheimer's Disease. Next, apply a model using machine learning (SVM) to this, and QSVM achieves the maximum accuracy of 99.01%. Experimental results indicate that the proposed Inception resnetv2 custom model and DenseNet201 perform better, with an accuracy of 98.12% and 95.68%, respectively. The recommended method can assist medical practitioners in early detection of Alzheimer's-related disease.

6.2 Limitation

A freely accessible MRI dataset was used for the study, that might not accurately reflect actual clinical data. Model performance may be impacted by the dataset's class imbalance, especially in the Moderate Demented group. The suggested approach does not use clinical, genetic, or population data for diagnosis; instead, it only uses MRI scans.

6.3 Future Work

To enhance model generality, future studies might concentrate on adding bigger and more varied MRI datasets gathered from other medical facilities. To improve diagnosis accuracy, advanced architectures for deep learning, explainable AI (XAI)

approaches, and multimodal data, including clinical records, DNA data, and cognitive evaluation scores, may be combined. Additionally, the recommended structure is capable of being implemented as a mobile or online based medical software to facilitate clinical decision-making and real-time Alzheimer's disease testing.

References

- [1] S. Sharma, A. Kumar, and R. Gupta, “Machine learning approaches for early Alzheimer’s disease detection using MRI data,” *Biomedical Signal Processing and Control*, vol. 86, pp. 105–118, 2023.
- [2] M. Rahman, S. Islam, and M. Hasan, “Support vector machine-based classification of Alzheimer’s disease from neuroimaging data,” *Expert Systems with Applications*, vol. 225, pp. 120–132, 2023.
- [3] Y. Wang, J. Li, and X. Zhang, “Random Forest model for Alzheimer’s disease prediction using brain MRI features,” *Healthcare Analytics*, vol. 4, pp. 100–112, 2024.
- [4] R. Islam and M. Ahmed, “K-nearest neighbor based cognitive impairment detection using MRI images,” *International Journal of Medical Informatics*, vol. 181, pp. 105–116, 2023.
- [5] T. Zhang, H. Chen, and L. Wang, “Artificial neural network framework for Alzheimer’s disease diagnosis,” *Computers in Biology and Medicine*, vol. 170, Art. no. 107891, 2024.
- [6] J. Lee, S. Kim, and H. Park, “Deep convolutional neural networks for Alzheimer’s disease classification using MRI scans,” *IEEE Access*, vol. 12, pp. 45678–45689, 2024.
- [7] P. Singh and K. Verma, “LSTM-based deep learning model for Alzheimer’s disease progression prediction,” *Neural Computing and Applications*, vol. 37, no. 2, pp. 1155–1168, 2025.
- [8] H. Chen, Y. Zhao, and X. Liu, “Hybrid CNN-SVM architecture for Alzheimer’s disease diagnosis using MRI images,” *Biomedical Signal Processing and Control*, vol. 91, Art. no. 105812, 2024.
- [9] Y. Wang, M. Zhou, and J. Sun, “Multimodal deep learning for accurate Alzheimer’s disease detection,” *Artificial Intelligence in Medicine*, vol. 148, Art. no. 102756, 2025.

- [10] B. Roy, S. Dutta, and P. Ghosh, “Explainable artificial intelligence framework for Alzheimer’s disease diagnosis,” *Knowledge-Based Systems*, vol. 294, Art. no. 111623, 2024.
- [11] A. Krizhevsky, I. Sutskever, and G. E. Hinton, “ImageNet classification with deep convolutional neural networks,” *Communications of the ACM*, vol. 60, no. 6, pp. 84–90, 2017.
- [12] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, “Densely connected convolutional networks,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI, USA, 2017, pp. 4700–4708.
- [13] J. Hu, L. Shen, and G. Sun, “Squeeze-and-excitation networks,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Salt Lake City, UT, USA, 2018, pp. 7132–7141.
- [14] C. Cortes and V. Vapnik, “Support-vector networks,” *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [15] M. Schuld and F. Petruccione, *Supervised Learning with Quantum Computers*. Cham, Switzerland: Springer, 2018.

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